THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $15\sqrt{3}$ m	$h = 15\tan 60^\circ = 15 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 15\sqrt{3}$ m
2	**(d)** $3\sqrt{3}$ m	Height $= 6\sin 60^\circ = 6 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 3\sqrt{3}$ m
3	**(d)** 12 m	Add the height of the eyes: $2 + 10$; Height of building $= 12$ m
4	**(d)** $10\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$; Height $= 10\sqrt{3}$ m (distances 30 m and 10 m)
5	**(c)** 12 m	Distance from the foot $= 4\cot 30^\circ = 4\sqrt{3}$ m; Original height of the tree $= 12$ m (that is $4 + 8$)
6	**(a)** $7 + 7\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 7\sqrt{3}$ m; Tower $= 7 + 7\sqrt{3}$ m
7	**(b)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
8	**(a)** $7\sqrt{3}$ m	$h = 21\tan 30^\circ = 21 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 7\sqrt{3}$ m
9	**(a)** $8\sqrt{3}$ m	Height $= 16\sin 60^\circ = 16 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 8\sqrt{3}$ m
10	**(b)** 18 m	Add the height of the eyes: $3 + 15$; Height of building $= 18$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Broken part (hypotenuse) $= \frac{6}{\sin 30^\circ} = 12$ m; Distance from the foot $= 6\cot 30^\circ = 6\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 6\sqrt{3}$ m (1).	1+1
12	The ladder is the hypotenuse, length 14 m; Base $= 14\cos 60^\circ = 14 \times \frac{1}{2}$ (1); Distance of the foot of the ladder from the wall $= 7$ m (1).	1+1
13	$\tan\theta = \frac{\text{rise}}{\text{run}} = \frac{10\sqrt{3}}{10}$; $\tan\theta = \sqrt{3}$ which is $\tan 60^\circ$ (1); $\theta = 60^\circ$ (1).	1+1
14	Broken part (hypotenuse) $= \frac{8}{\sin 30^\circ} = 16$ m; Distance from the foot $= 8\cot 30^\circ = 8\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 8\sqrt{3}$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 20 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 10\sqrt{3}$ m (distances 30 m and 10 m) (1).	1+1+1
16	Horizontal distance $d = 5\cot 45^\circ = 5$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 5\sqrt{3}$ m (1); Tower $= 5 + 5\sqrt{3}$ m (1).	1+1+1
17	Distances from the foot: $h\cot 30^\circ = 75\sqrt{3}$ m (farther) and $h\cot 45^\circ = 75$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 75\sqrt{3} - 75$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 34 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 34$ gives $h = 34 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 17\sqrt{3}$ m (distances 51 m and 17 m) (1).	1+1+1
18	Horizontal distance $d = 8\cot 45^\circ = 8$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 8\sqrt{3}$ m (1); Tower $= 8 + 8\sqrt{3}$ m (1).	1+1+1
	OR: Distances from the foot: $h\cot 30^\circ = 60\sqrt{3}$ m (farther) and $h\cot 45^\circ = 60$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 60\sqrt{3} - 60$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 80 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 80$ gives $h = 20\sqrt{3}$ (2); Height $20\sqrt{3}$ m; distances 20 m and 60 m (1).	2+2+1

	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 2 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 2$ (2); $p = \frac{2}{\sqrt{3}-1} = 1 + \sqrt{3}$ m (1).	2+2+1
20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 120 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 120$ gives $h = 30\sqrt{3}$ (2); Height $30\sqrt{3}$ m; distances 30 m and 90 m (1).	2+2+1
	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 4 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 4$ (2); $p = \frac{4}{\sqrt{3}-1} = 2 + 2\sqrt{3}$ m (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $15\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 30$ gives $h = 30 \div (\frac{2\sqrt{3}}{3})$; Height $= 15\sqrt{3}$ m (distances 45 m and 15 m)
2	**(d)** 15 m	Distance from the foot $= 5\cot 30^\circ = 5\sqrt{3}$ m; Original height of the tree $= 15$ m (that is $5 + 10$)
3	**(d)** $10 + 10\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 10\sqrt{3}$ m; Tower $= 10 + 10\sqrt{3}$ m
4	**(c)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
5	**(c)** 40 m	$h = 40\tan 45^\circ = 40 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 40$ m
6	**(c)** $12\sqrt{3}$ m	Height $= 24\sin 60^\circ = 24 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 12\sqrt{3}$ m
7	**(a)** $2 + 12\sqrt{3}$ m	Add the height of the eyes: $2 + 12\sqrt{3}$; Height of building $= 2 + 12\sqrt{3}$ m
8	**(d)** $20\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 40$ gives $h = 40 \div (\frac{2\sqrt{3}}{3})$; Height $= 20\sqrt{3}$ m (distances 60 m and 20 m)
9	**(d)** 24 m	Distance from the foot $= 8\cot 30^\circ = 8\sqrt{3}$ m; Original height of the tree $= 24$ m (that is $8 + 16$)
10	**(d)** $14 + 14\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 14\sqrt{3}$ m; Tower $= 14 + 14\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The ladder is the hypotenuse, length 22 m; Base $= 22\cos 60^\circ = 22 \times \frac{1}{2}$ (1); Distance of the foot of the ladder from the wall $= 11$ m (1).	1+1
12	$\tan\theta = \frac{\text{rise}}{\text{run}} = \frac{12\sqrt{3}}{12}$; $\tan\theta = \sqrt{3}$ which is $\tan 60^\circ$ (1); $\theta = 60^\circ$ (1).	1+1
13	Broken part (hypotenuse) $= \frac{10}{\sin 30^\circ} = 20$ m; Distance from the foot $= 10\cot 30^\circ = 10\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 10\sqrt{3}$ m (1).	1+1
14	The ladder is the hypotenuse, length 28 m; Base $= 28\cos 60^\circ = 28 \times \frac{1}{2}$ (1); Distance of the foot of the ladder from the wall $= 14$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 50 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 50$ gives $h = 50 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 25\sqrt{3}$ m (distances 75 m and 25 m) (1).	1+1+1
16	Horizontal distance $d = 9\cot 45^\circ = 9$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 9\sqrt{3}$ m (1); Tower $= 9 + 9\sqrt{3}$ m (1).	1+1+1
17	Distances from the foot: $h\cot 30^\circ = 90\sqrt{3}$ m (farther) and $h\cot 45^\circ = 90$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 90\sqrt{3} - 90$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 40 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 40$ gives $h = 40 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 20\sqrt{3}$ m (distances 60 m and 20 m) (1).	1+1+1
18	Horizontal distance $d = 15\cot 45^\circ = 15$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 15\sqrt{3}$ m (1); Tower $= 15 + 15\sqrt{3}$ m (1).	1+1+1
	OR: Distances from the foot: $h\cot 30^\circ = 45\sqrt{3}$ m (farther) and $h\cot 45^\circ = 45$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 45\sqrt{3} - 45$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 160 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 160$ gives $h = 40\sqrt{3}$ (2); Height $40\sqrt{3}$ m; distances 40 m and 120 m (1).	2+2+1

	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 6 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 6$ (2); $p = \frac{6}{\sqrt{3}-1} = 3 + 3\sqrt{3}$ m (1).	2+2+1
20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 40 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 40$ gives $h = 10\sqrt{3}$ (2); Height $10\sqrt{3}$ m; distances 10 m and 30 m (1).	2+2+1
	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 8 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 8$ (2); $p = \frac{8}{\sqrt{3}-1} = 4 + 4\sqrt{3}$ m (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
2	**(d)** $2\sqrt{3}$ m	$h = 6\tan 30^\circ = 6 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 2\sqrt{3}$ m
3	**(a)** $15\sqrt{3}$ m	Height $= 30\sin 60^\circ = 30 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 15\sqrt{3}$ m
4	**(c)** 21 m	Add the height of the eyes: $1 + 20$; Height of building $= 21$ m
5	**(c)** $30\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 60$ gives $h = 60 \div (\frac{2\sqrt{3}}{3})$; Height $= 30\sqrt{3}$ m (distances 90 m and 30 m)
6	**(a)** 27 m	Distance from the foot $= 9\cot 30^\circ = 9\sqrt{3}$ m; Original height of the tree $= 27$ m (that is $9 + 18$)
7	**(b)** $12 + 12\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 12\sqrt{3}$ m; Tower $= 12 + 12\sqrt{3}$ m
8	**(b)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
9	**(b)** $35\sqrt{3}$ m	$h = 35\tan 60^\circ = 35 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 35\sqrt{3}$ m
10	**(b)** $9\sqrt{3}$ m	Height $= 18\sin 60^\circ = 18 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 9\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\tan\theta = \frac{\text{rise}}{\text{run}} = \frac{14\sqrt{3}}{14}$; $\tan\theta = \sqrt{3}$ which is $\tan 60^\circ$ (1); $\theta = 60^\circ$ (1).	1+1
12	Broken part (hypotenuse) $= \frac{15}{\sin 30^\circ} = 30$ m; Distance from the foot $= 15\cot 30^\circ = 15\sqrt{3}$ m (1); Distance from the foot of the tree to the point where its top touches the ground $= 15\sqrt{3}$ m (1).	1+1
13	The ladder is the hypotenuse, length 36 m; Base $= 36\cos 60^\circ = 36 \times \frac{1}{2}$ (1); Distance of the foot of the ladder from the wall $= 18$ m (1).	1+1
14	$\tan\theta = \frac{\text{rise}}{\text{run}} = \frac{16\sqrt{3}}{16}$; $\tan\theta = \sqrt{3}$ which is $\tan 60^\circ$ (1); $\theta = 60^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 24 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 24$ gives $h = 24 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 12\sqrt{3}$ m (distances 36 m and 12 m) (1).	1+1+1
16	Horizontal distance $d = 16\cot 45^\circ = 16$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 16\sqrt{3}$ m (1); Tower $= 16 + 16\sqrt{3}$ m (1).	1+1+1
17	Distances from the foot: $h\cot 30^\circ = 100\sqrt{3}$ m (farther) and $h\cot 45^\circ = 100$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 100\sqrt{3} - 100$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 60 m (1); $h(\cot 30^\circ - \cot 60^\circ) = 60$ gives $h = 60 \div (\frac{2\sqrt{3}}{3})$ (1); Height $= 30\sqrt{3}$ m (distances 90 m and 30 m) (1).	1+1+1
18	Horizontal distance $d = 20\cot 45^\circ = 20$ m (from the depression of the foot) (1); Part of the tower above the top of the building $= d\tan 60^\circ = 20\sqrt{3}$ m (1); Tower $= 20 + 20\sqrt{3}$ m (1).	1+1+1
	OR: Distances from the foot: $h\cot 30^\circ = 30\sqrt{3}$ m (farther) and $h\cot 45^\circ = 30$ m (nearer) (1); Both objects are on the same side, so the distances are subtracted (1); Distance $= 30\sqrt{3} - 30$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 200 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 200$ gives $h = 50\sqrt{3}$ (2); Height $50\sqrt{3}$ m; distances 50 m and 150 m (1).	2+2+1
	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p + 10 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 10$ (2); $p = \frac{10}{\sqrt{3}-1} = 5 + 5\sqrt{3}$ m (1).	2+2+1

20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 240 (2); $h\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right)=240$ gives $h=60\sqrt{3}$ (2); Height $60\sqrt{3}$ m; distances 60 m and 180 m (1).	2+2+1
	OR: Let pedestal $= p$; distance from the point $= p\cot 45^\circ = p$ (2); $p+12 = p\tan 60^\circ$ gives $p(\sqrt{3}-1)=12$ (2); $p=\frac{12}{\sqrt{3}-1}=6+6\sqrt{3}$ m (1).	2+2+1

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- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
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- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
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